
Education

University of Toronto Scarborough | BSc in Statistics (Quantitative Finance)

Expected completion: 12/2027

Relevant coursework: Probability, Statistics, Stochastic Processes, Linear Programming and Optimization, Numerical Algorithms for Computational Mathematics, Derivative Markets, Multivariable Calculus, Linear Algebra I/II, Differential Equations, Data Structures, Python and C Programming

Skills

Quantitative & Modelling: Options pricing (Black-Scholes), Implied volatility, Numerical root-finding, Least-squares fitting, Statistical modelling, Time-series analysis, Backtesting

Programming & Data: Python (NumPy, pandas, SciPy), C, SQL, Power BI, Data cleaning, Data visualisation

Projects

[GitHub](#)

Implied Volatility Surfaces and the Variance Risk Premium | Python, SciPy, SQL

- Implemented Black-Scholes pricing on the forward and derived analytic vega, validating both against put-call parity, monotonicity in volatility, payoff convergence at expiry, and a finite-difference check.
- Querying WRDS/OptionMetrics via SQL to build a filtered sample of listed index option quotes, logging rows removed at each filter stage to keep the sample defensible.
- Recovering implied volatility by Brent inversion and fitting SVI surfaces per maturity, testing fitted surfaces for butterfly and calendar arbitrage violations.
- Redesigning the cross-market comparison mid-build after verifying the data subscription lacked Canadian coverage, substituting NYSE-listed EWC options and documenting the limitations. (*In progress*)

Quant Trading Strategy Backtest | Python

- Built a backtesting framework for indicator-based strategies on TSLA daily data, with transaction costs, position sizing, and volatility targeting applied at the signal level.
- Tested across market regimes and found the apparent edge did not survive realistic cost assumptions; reported the negative result rather than tuning parameters until it looked profitable.

Music Data Structures Project | C

- Built a music-processing program in C using binary search trees and linked structures to store, transform, and play back note sequences from parsed files.
- Implemented insertion, search, deletion, traversal, song reversal, and harmonisation with manual memory management throughout.

Ingredient Networks Graph Project | C

- Modelled ingredient relationships as an adjacency-matrix graph in C, with recursive k-hop traversal to find indirect connections and exclusion filtering for dietary constraints.
- Ranked substitution candidates from the network, validated on graphs up to 400 nodes.

Please Turn Over.

Work Experience

Finance Intern | Heineken Canada

Toronto, Canada | 05/2025 - 12/2025

- Co-developed Python-based automation to transform and standardize Excel data files, reducing manual processing and improving reporting reliability.
- Prepared and delivered weekly finance presentations to cross-functional stakeholders, presenting Power BI dashboards and synthesizing key business performance trends across sales, pricing, discounting, and channel mix to support planning and decision-making.
- Extracted, cleaned, and analyzed large sales and finance datasets using Excel and Power BI, building dashboards to track monthly sales volumes (hectolitres) across retail, LCBO, grocery, and other channels.
- Performed time-series and comparative analysis (R1, R3, R6, R12) on brand performance, competitor pricing, and discounting using pivot tables and structured data models to support forecasting and strategy discussions.
- Analyzed seasonality trends in sales volumes and collaborated with finance and sales stakeholders to translate data insights into actionable inputs for planning and cash-flow optimization for the Fall-Winter season 2025.

Junior Financial Analyst | Teesta Solar Ltd.

Dhaka, Bangladesh | 05/2024 - 08/2024

- Built a 20-year revenue model in Excel for a solar power project, incorporating accounting assumptions, scenario analysis, and return projections.
- Developed a DCF model for a 300 MW demo solar plant, including sensitivity analysis and financing assumptions to evaluate project feasibility and investment returns.
- Supported investment analysis by assessing projected cash flows, cost drivers, and return scenarios for management review, and assisted in developing KPI dashboards to monitor energy output, operating efficiency, and cost performance.

Extracurricular Activities

Finance & Marketing Director | Data Science and Statistics Society - UofT

Toronto, Canada | 09/2024 - 05/2026

- Hosted the DS3 Datathon 2025, overseeing budget allocations, sponsor coordination, and financial documentation, while tracking KPI performance across all event functions.